

Custom Converters

These are special type of Functions that convert HTML tags into the appropriate HTML.

There are standard converters (called Node Converters) defined in the system.

Node Converter	ExecutionOrderIndex	Description	HTML Layout Editor	HTML Editor (HTML View)
Dom-bind Converter	5	Adds data binding support.		✓
Preview Converter	10	Generates "preview" attributes (only for preview).	✓	
Remover Script Converter	20	Removes scripts if the "Include JavaScript" option is disabled (only for preview).	✓	
Iterate Converter	30	Adds <i>data-flx-iterate</i> (<i>data-flx-iterate-children</i>) attribute support.	✓	✓
Boolean Attributes Converter	40	Adds <i>data-flx-attr</i> attribute support.	✓	✓
Literal Converter	50	Adds literals support.	✓	✓
If Converter	55	Adds conditionals support.	✓	✓
Screen Interface Inputs Converter	60	Allows for parsing Screen Interface Input values.	✓	✓
Filters Converter	70	Adds <i>data-flx-filter</i> attribute support.	✓	✓
Step Grid Converter	80	Adds Step Grid, <i>data-flx-stepgrid</i> attribute support.	✓	
Standard Inputs Converter	90	Adds standard Inputs, <i>data-flx-input</i> attribute support.	✓	
Standard Outputs Converter	100	Adds standard Outputs, <i>data-flx-output</i> attribute support.	✓	
Sub-DFC Converter	110	Adds Sub-DFC, <i>data-flx-subdfc</i> (previously <i>data-flx-subop</i>) attribute support.	✓	
Business Controls Converter	120	Adds Business Control, <i>data-flx-bc</i> attribute support.	✓	
Buttons Tabs Converter	127	Adds Actions, <i>data-flx-button</i> , <i>data-flx-button-disabled</i> , <i>data-flx-taband</i> and <i>data-flx-tab-text</i> attribute support.		✓
Action Converter	128	Adds <i>data-flx-action</i> attribute support.		✓
Dynamic Bindings Converter	129	Adds <i>data-flx-dynamic-bind</i> attribute support.	✓	✓
Output Bindings Converter	130	Adds <i>data-flx-bind</i> attribute support.	✓	✓
Screen Context Converter	140	Adds <i>\$Context</i> and <i>\$View</i> variables script support	✓	✓

		(JavaScript).		
ScreenInitConverter	160	Is responsible for initializing \$Context JS variable.	✓	✓
HTML Controls Preserving Converter	170	Adds support for preserving HTML control values in case of server error.	✓	✓
Key Tag Converter	180	Adds <i>data-flx-key</i> attribute support	✓	✓

To Add a Custom Converter

1. Create an assembly (that needs to be signed) containing classes implementing the *FlexNetProcess Builder2.BusinessRules.ScreenModel.Converters.INodeConverter* interface (*FlexNet.ProcessBuilder2.BusinessRules.ScreenModel.dll*).
2. Copy a DLL file to GAC and Program Files\Dassault Systemes\DELMIA Apriso 2025\WebSite\Downloads\PB2 (on the server).
3. Add the converter assembly name to the "ExtensionAssemblies" key located in the "ProcessBuilder" section of the Central Configuration file (for details, see [Central Configuration Documentation](#)). Its value is a list of assemblies (separated by '|') containing converter definitions (classes that derive from the *INodeConverter*). See the example below:

```
<add key="ExtensionAssemblies" value="MyCompany.CustomNodeConverter, Version=1.0, Culture=neutral, PublicKeyToken=xxxxxxxxxxxx | MyCompany.CustomNodeConverter2, Version=1.5, Culture=neutral, PublicKeyToken=xxxxxxxxxxxx" />
```

4. Rebuild **ClickOnce** and restart **DELMIA Apriso Services/IIS**.

Node Converters will be automatically loaded based on the implemented interface. Additionally, each Converter contains an index of execution, which sets all of the converters in a specified order and allows the custom converters to be executed in a specified moment (before, after, or between the DELMIA Apriso converters).

To Add a JavaScript Custom Converter

A JavaScript converter can be used in [HTML View](#).

1. Create a project containing classes that implement *FlexNetProcess Builder2.BusinessRules.ScreenModel.Converters.INodeConverter* interface (*FlexNet.ProcessBuilder2.BusinessRules.ScreenModel.dll*). Example:

```
public class CustomConverter : INodeConverter
{
    public ConverterScope Scope { get { return ConverterScope.ClientRuntime; } }
    public string Filter { get { return "data-flx-custom" } }
    public string JsConverterPath { get { return @"converters/CustomConverter.js"; } }
}

//rest of the interface
```

2. Generate a signed assembly.
3. Create a converter in JavaScript. Example:

```

var Converters;
(function(Converters) {
    var CustomConverter = (function() {
        function CustomConverter() {
        }
        CustomConverter.prototype.parse = function (html, params) {
        return new Promise(function (res, reject) {
            try {
                var found = html.querySelectorAll("[data-flx-custom]");
                if (found.length === 0){
                    return res(html);
                }else{
                    //
                    // your code here
                    //
                    return res(html);
                }
            }
            catch (err) {
                reject(err);
            }
        });
    });
    return CustomConverter;
})();
Converters.CustomConverter = CustomConverter;
//200 - Execution Order Index
Converters.ConverterManager.addConverter(200, new CustomConverter());
})(Converters || (Converters = {}));

```

Make sure that JS file is valid and does not cause any errors.

4. Copy the DLL file to the Global Assembly Cache using **gacutil.exe** located in <drive>\Program Files\Dassault Systemes\DELMIA Apriso 2025\Setup\Tools
In command line type: gacutil -i "<path to the assembly file>".
5. Copy the DLL file to <drive>\Program Files\Dassault Systemes\DELMIA Apriso 2025\WebSite\Downloads\PB2
6. Copy the JS file to <drive>\Program Files\Dassault Systemes\DELMIA Apriso 2025\WebSite\Portal\Scripts\converters
7. Add the converter assembly name to the "ExtensionAssemblies" key located in the "ProcessBuilder" section of the Central Configuration file (for details, see [Central Configuration Documentation](#)). Example:

```

<add key="ExtensionAssemblies" value="CustomConverter, Version=1.0.0.0,
Culture=neutral, PublicKeyToken=xxxxxxxxxxxx" />

```

8. Rebuild ClickOnce (run the **Publish Apriso Process Builder via ClickOnce** batch file located in <drive>\Program Files\Dassault Systemes\DELMIA Apriso 2025\WebSite\Downloads\ClickOnce Tools) and restart **DELMIA Apriso Services/IIS**.